Day 1: Foundations of Relativistic Kinematics & Experimental HEP

14-Day Hands-on Course on Heavy-Ion Phenomenology

Lecture

NIT-J,HEP-119

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Syllabus: 14-Day Hands-on Course Layout

Part I: Relativistic Kinematics

- **Day 1: Experimental HEP, Boosts & Rapidity**
- Day 2: Kinematics I: 4-vectors & CMS energy
- Day 3: Kinematics II: Rapidity & Pseudorapidity
- Day 4: Kinematics III: Invariant Yield & Phase Space

Part II: AMPT Physics & Mechanics

- Day 5: AMPT Evolution: HIJING, ZPC, Hadronization, ART
- Day 6: AMPT Data format: event headers & parsing
- Day 7: Single-Particle Observables: Multiplicity & p_T

Part III: Thermodynamics & Collectivity

- Day 8: Transverse Spectra, Identified Particles & m_T Scaling
- Day 9: Centrality & Geometry: Glauber vs AMPT
- Day 10: Collective Flow: Elliptic Flow (v_2)
- Day 11: Medium Modification: ZPC Knobs & Mean Free Path

Part IV: Advanced Scans & Pipeline

- Day 12: Energy Scan: Beam Energy Scan (STAR at RHIC)
- Day 13: Advanced Analysis
- Day 14: Wrap-Up: Reproducible HEP Data Pipeline

Why Experimental High Energy Physics? Probing the Extremes

The QGP Phase & Early Universe

- We collide heavy nuclei at ultra-relativistic velocities to recreate the conditions of the early universe (10^{-6} seconds after the Big Bang).
- **The Quark-Gluon Plasma (QGP):** A state of deconfined, color-charged quarks and gluons.

The QCD Confinement Challenge

- Hadrons are held together by strong force (QCD). Because of color confinement, we cannot isolate individual quarks/gluons directly.
- **Solution:** We must analyze final-state hadron sprays to infer the plasma's thermodynamic and transport properties.

Collision Energetics: Why Accelerators?

Resolving Matter at Sub-Femtometer Scales

- Accelerators provide high center-of-mass energies ($\sqrt{s}$) to satisfy mass-energy equivalence ($E = mc^2$).
- Short de Broglie wavelengths probe smaller structures: $\lambda = h/p$.

Center-of-Mass Energy Formulas

- **Collider Advantage:** In a collider ($E_{beam1} = E_{beam2} = E$), center-of-mass energy scales linearly:

$$\sqrt{s_{\text{collider}}} \approx 2E$$

- **Fixed Target Limit:** In a fixed-target collision (beam energy E on target nucleon mass m_N), energy scales only as the square root:

$$\sqrt{s_{\text{fixed}}} \approx \sqrt{2m_N E}$$

Fixed-Target vs. Collider Geometry

Center-of-Mass Energetics Comparison

Let us calculate the beam energy required to match LHC and RHIC center-of-mass energies in a fixed-target configuration:

System	**Collider $\sqrt{s_{NN}}$**	**Fixed-Target Equivalent E_{beam}**	**Kinematic Advantage**
SPS ($p + p$)	17.3 GeV	158 GeV	Baseline
RHIC ($Au + Au$)	200 GeV	21.3 TeV	100$\times$ Beam boost
LHC ($Pb + Pb$)	5.02 TeV	13.5 PeV (13,500 TeV)	Outside classical accelerator tech

Interactive Simulation

Open our custom simulator at `animations/collision_animation.html` to interactively run collisions and observe the center-of-mass energy transformations.

Specialization by Design: Belle II vs. LHCb

Belle II (KEKB Asymmetric Collider)

- **Beams:** 8.0 GeV e^- on 3.5 GeV e^+.
- **Design:** Collides asymmetric energies to provide a longitudinal Lorentz boost to the center of mass.
- **Physics:** Allows precise decay-length measurements of short-lived B-mesons ($B \to J/\psi K_S^0$) to study CP violation.

LHCb (LHC Forward Spectrometer)

- **Beams:** Symmetric proton-proton at 13.6 TeV.
- **Design:** Forward single-arm cone spectrometer ($2 < \eta < 5$).
- **Physics:** At LHC energies, $b\bar{b}$ pairs are produced at very forward/backward angles. Placed in the forward direction to capture these heavy flavor hadrons.

Specialization by Design: STAR vs. CMS vs. ALICE

ALICE (LHC)

QGP Specialized Tracking

- Optimized for extreme track density ($Pb + Pb$ collisions).
- Extremely low material budget and magnetic field (0.5 T) to track and identify very low-p_T hadrons.

STAR (RHIC)

Hadron Spectroscopy

- Massive Time Projection Chamber (TPC) optimized for particle identification, centrality scans, and collective flow physics at RHIC energies.

CMS (LHC)

High-p_T & High-Luminosity

- Built around a massive 3.8 T superconducting solenoid.
- High-granularity silicon trackers and deep EM/Hadronic calorimeters designed to capture high-p_T jets, leptons, and Higgs candidates.

Layered "Onion" Detector Architecture

Concentric Cylindrical Configurations

To track and reconstruct every single final-state particle, modern cylindrical colliders use concentric detector layers centered around the beam axis:

1. **Vertex Detectors (Inner Silicon):** Capture prompt secondary decays close to the interaction point.
2. **Central Trackers (TPC / Drift Chambers):** Map trajectories under magnetic fields to resolve momenta.
3. **Particle Identification (TOF / RICH):** Separate pions, kaons, and protons.
4. **Electromagnetic Calorimeters (EMCal):** Capture electron and photon energy.
5. **Hadronic Calorimeters (HCal):** Absorb hadronic shower energy.
6. **Muon Spectrometer:** Outermost chambers that capture highly penetrating muons.

Tracking & Curvature in Time Projection Chambers (TPC)

TPC Operation Principles

- Gas-filled drift chamber placed inside high electric and magnetic fields.
- Traversing charged particles ionize gas atoms, producing electrons that drift towards anodes.
- Reconstructs 3D tracks from anode pixels (X, Y) and drift times (Z).

TPC Physics Outputs

- **Momentum Measurement:** Curved tracks in magnetic field B yield momentum p:

$$p = 0.3 \cdot B \cdot R$$

  where R is the curvature radius.
- **Energy Loss (dE/dx):** Measures the rate of specific ionization loss to identify particles via Bethe-Bloch curves at low momentum.

Particle Identification (PID): Time-of-Flight vs. RICH

Time-of-Flight (TOF) Systems

- Measures the time difference Δt between collision vertex and detector impact.
- **TOF Resolution:** High-precision Multigap Resistive Plate Chambers (MRPC) achieve resolutions of $\sigma_t \sim 50$–80 ps.
- Separates pions, kaons, and protons at low-intermediate transverse momenta.

Ring Imaging Cherenkov (RICH)

- Relies on Cherenkov radiation emitted when a particle's speed exceeds speed of light in medium: $\beta > 1/n$.
- Measures Cherenkov angle θ_C:

$$\cos\theta_C = \frac{1}{n\beta}$$

- Projects rings onto photon detectors, separating high-momentum species.

Energy Capture: Calorimeters & Muon Spectrometers

Calorimetry Layers

- **Electromagnetic Calorimeter (EMCal):** Uses high-Z absorbers (e.g. Lead-Tungstate crystals) to trigger Bremsstrahlung showers, capturing all $e^{\pm}$ and γ energy.
- **Hadronic Calorimeter (HCal):** Absorbs strongly interacting hadrons ($\pi^{\pm}, K^{\pm}, p, n$) through inelastic nuclear interactions.

Muon Spectrometers

- Muons are minimum ionizing particles (MIPs) and do not trigger hadronic or electromagnetic showers.
- They penetrate through thick iron shielding.
- Placed outermost to record clean muon tracks (crucial for vector meson decay $J/\psi \to \mu^+\mu^-$).

Why AMPT? The Dynamical Multi-Phase Transport

The Physics Motivation

- Detectors only record the asymptotic hadronic state after **kinetic freeze-out**.
- **AMPT (A Multi-Phase Transport)** is a dynamical Monte Carlo transport model that reconstructs the complete collision history:

  Initial Geometry (HIJING) → Parton Cascade (ZPC) → Hadronization (Coalescence) → Had

Course Outcomes for Students

By the end of this 14-day course, you will be able to:

1. Stream and parse raw simulated heavy-ion data.
2. Code relativistic kinematics and flow observables (v_2) from scratch.
3. Build full reproducible data analysis pipelines.

Relativistic Kinematics: Collinear Boosts (Sahoo pg. 115-116)

Successive Boosts along X-axis

Consider three collinear inertial frames of reference $S \xrightarrow{v} S' \xrightarrow{u} S''$ moving along the positive X-axis (Sahoo Fig. 5.6):

Boost 1: $S \to S'$ with velocity v ($\beta_1 = v/c$):

$$x'^0 = \gamma_v(x^0 - \beta_1 x^1)$$

$$x'^1 = \gamma_v(x^1 - \beta_1 x^0)$$

where $\gamma_v = \frac{1}{\sqrt{1-\beta_1^2}}$

Boost 2: $S' \to S''$ with velocity u ($\beta_2 = u/c$):

$$x''^0 = \gamma_u(x'^0 - \beta_2 x'^1)$$

$$x''^1 = \gamma_u(x'^1 - \beta_2 x'^0)$$

where $\gamma_u = \frac{1}{\sqrt{1-\beta_2^2}}$

Relativistic Velocity Addition

Combined Single Boost $S \to S''$ with velocity w

Substituting the coordinate equations from Boost 1 into Boost 2 yields the effective relativistic velocity boost factor (Sahoo Eq. 5.41):

$$w = \frac{v + u}{1 + \frac{vu}{c^2}}$$

And the total combined Lorentz factor becomes (Sahoo Eq. 5.39):

$$\gamma_w = \gamma_u \gamma_v \left(1 + \frac{uv}{c^2}\right)$$

Interactive Relativistic Demonstrations

Open in your browser:

- `animations/rapidity_boost.html` (Linear rapidity boosts & additivity)
- `animations/01_velocity_saturation_comparison.html` (Velocity saturation vs

Rigorous Limits of Velocity Addition

Universal Speed Limit

This non-linearity guarantees that the speed of light c is the absolute speed limit in the universe:

$$\text{If } v \to c \quad \text{or} \quad u \to c \quad \Rightarrow \quad w \to c$$

Classical vs. Relativistic Velocity Saturation

In Problem 1 of the hands-on lab, you will successively add collinear boosts of $0.1c$ up to 20 times. - **Classical sum** grows linearly to $2.0c$. - **Relativistic sum** saturates at $0.964c$, asymptotically approaching $1.0c$, as shown in your saved plot `day01_problem1_velocity_rapidity.png`.

Rapidity Definition (Sahoo pg. 117-118)

Hyperbolic Transformations in Minkowski Space

Because velocity addition is highly non-linear, measuring spectrum shapes and calculating transformations across moving frames in colliders is mathematically complex. We define a new hyperbolic angle called **Rapidity** (y):

$$y = \tanh^{-1}(\beta) = \frac{1}{2}\ln\left(\frac{1+\beta}{1-\beta}\right)$$

Hyperbolic mappings transform the velocity parameter:

$$\beta = \tanh(y), \quad \gamma = \cosh(y), \quad \gamma\beta = \sinh(y)$$

Proof of Rapidity Additivity: Mathematical Derivation

Hyperbolic Angle Transformations (Sahoo pg. 117-118)

Let $y_1 = \tanh^{-1}(u/c)$ and $y_2 = \tanh^{-1}(v/c)$ represent two collinear boosts (taking $c = 1$):

$$w = \frac{u+v}{1+uv} = \frac{\tanh y_1 + \tanh y_2}{1 + \tanh y_1 \tanh y_2}$$

The Hyperbolic Tangent Identity

Recalling the standard hyperbolic tangent angle-addition identity:

$$\tanh(y_1 + y_2) = \frac{\tanh y_1 + \tanh y_2}{1 + \tanh y_1 \tanh y_2}$$

Which gives us the combined velocity parameter:

$$w = \tanh(y_1 + y_2)$$

Rapidity Additivity: Physical Consequences

Proving Linear Additivity (Sahoo Eq. 5.43)

Since the effective relativistic velocity boost is $w = \tanh(y_{\text{tot}})$, substituting our derivation yields:

$$\tanh(y_{\text{tot}}) = \tanh(y_1 + y_2) \quad \Rightarrow \quad y_{\text{tot}} = y_1 + y_2$$

This simple, elegant relation proves that collinear boosts combine linearly under rapidity!

Takeaway: Natural Collider Spectroscopy Variable

* **Velocity Addition** is highly non-linear, making cross-frame spectrum transformations mathematically complex. * **Rapidity Addition** is linear. Boosts along the beam axis simply shift the rapidity distribution by a constant ($y \to y + y_{\text{boost}}$), keeping the shape of the invariant yield distribution dN/dy perfectly invariant!

Kinematics in the Center-of-Momentum System (Sahoo pg. 113-114)

Asymmetric 2-Body Kinematics

Let p_1 and p_2 be the momentum 4-vectors of two colliding particles with masses m_1 and m_2. Let $P = p_1 + p_2$ be the total center-of-momentum 4-vector, with total invariant mass squared:

$$P^2 = (p_1 + p_2)^2 = M^2 = s$$

Individual CMS Energies (Sahoo Eq. 5.29 & 5.30)

Transforming to the center-of-momentum frame ($P = (\sqrt{s}, \vec{0})$) yields:

$$E_1^* = \frac{s + m_1^2 - m_2^2}{2\sqrt{s}}$$

$$E_2^* = \frac{s - m_1^2 + m_2^2}{2\sqrt{s}}$$

CMS Momentum & Relative Velocity

CMS Momentum p^* (Sahoo Eq. 5.32)

By using the invariant mass relation $E_1^{*2} - |\vec{p}^*|^2 = m_1^2$, we solve for the exact center-of-momentum three-momentum magnitude:

$$|\vec{p}^*|^2 = \frac{s^2 - 2s(m_1^2 + m_2^2) + (m_1^2 - m_2^2)^2}{4s}$$

Note that $E_1^* + E_2^* = \sqrt{s}$ and the particles move in opposite directions with equal momentum magnitude: $\vec{p}_1^* = -\vec{p}_2^*$.

Relative CMS Velocity (Sahoo Eq. 5.33)

The relative CMS velocity parameter is calculated as:

$$v_{12}^* = \left(\frac{|\vec{p}^*|}{E_1^*}\right)^2$$

Real Data: Symmetric vs. Asymmetric CMS Kinematics

Computed Metrics from Problem 2

These exact kinematic formulas yield the following physical values for symmetric and asymmetric collider configurations:

- **Symmetric $p + p$ at ALICE ($\sqrt{s} = 13.6$ TeV):**

$$E_1^* = E_2^* = 6800.000 \text{ GeV}, \quad p^* = 6800.000 \text{ GeV/c}$$

- **Symmetric $Au + Au$ at STAR ($\sqrt{s_{NN}} = 200$ GeV):**

$$E^*_{\text{nucleon}} = 100.000 \text{ GeV}, \quad p^* = 99.996 \text{ GeV/c}$$

- **Asymmetric $p + Pb$ at LHC ($\sqrt{s_{NN}} = 8.16$ TeV):**

$$E_p^* = 4077.666 \text{ GeV}, \quad E_{Pb}^* = 4082.334 \text{ GeV}, \quad p^* = 4077.666 \text{ GeV/c}$$

Spacetime Evolution Stages of Heavy-Ion Collisions

Dynamical Stages (Sahoo Fig. 5.7)

Relativistic collisions evolve through highly distinct physical epochs:

1. **Initial Collision (Glasma Phase):** Extreme gluon fields at $t \approx 0$ fm/c.
2. **Pre-Equilibrium & Thermalization:** Rapid interaction creates a deconfined system.
3. **Viscous Relativistic Hydrodynamics:** Strongly coupled QGP expands as a fluid ($0.5 < t < 10$ fm/c).
4. **Hadronization:** Fluid cools to a hadron gas at critical temperature $T_c \approx 155$ MeV.
5. **Hadronic Gas Transport (ART):** Hadrons collide, scatter, and freeze out.

Chronological Spacetime Diagram

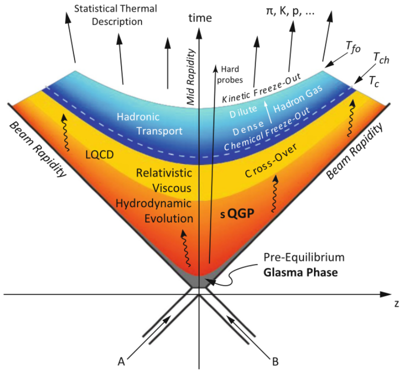


Fig. 5.7 Description of heavy-ion collisions in one space (z) and one time (t) dimension, with two nuclei colliding at ($t = 0, z = 0$). The formation of a very hot and dense fireball is envisaged, which follows a complex and dynamical spacetime evolution involving different quanta of the system at different time scales. The fireball passes through a pre-equilibrium phase, followed by the formation of a QGP and a cross-over phase transition to a hadron gas. This transition is marked by a critical temperature, T_c after which the chemical freeze-out and kinetic freeze-out of the system happen at temperatures, T_{ch} and T_{kin}, respectively. Final state secondary particles free-stream after

Coordinate Mappings: Light-Cone Momentum Variables

Definitions (Sahoo pg. 119-120)

For ultra-relativistic collinear longitudinal processes, we define light-cone momentum variables:

$$p_+ = E + p_z, \quad p_- = E - p_z$$

Which are highly advantageous since they scale simply under longitudinal boosts:

$$p'_+ = e^{-y_{boost}} p_+, \quad p'_- = e^{y_{boost}} p_-$$

Relation to Standard Rapidity

Using these variables, standard momentum rapidity y can be expressed as:

$$y = \frac{1}{2} \ln \left(\frac{p_+}{p_-} \right)$$

In Problem 4, you verify that this formulation matches standard rapidity with exact

Coordinate Mappings: Spacetime Rapidity η_s

Longitudinal Spacetime Scaling (Sahoo Eq. 5.44)

To represent the spatial coordinates (z, t) of particle freezeouts in an expanding QGP medium, we define the **Spacetime Rapidity** (η_s):

$$\eta_s = \frac{1}{2} \ln\left(\frac{t+z}{t-z}\right)$$

Analogously, we define the proper lifetime τ:

$$\tau = \sqrt{t^2 - z^2}$$

Bjorken Longitudinal Scaling Model

In ultra-relativistic collisions, spatial position and momentum become highly correlated: particles with large forward velocities end up at large forward spatial coordinates.

Real AMPT Correlation: Spacetime vs. Momentum Rapidity

Validation on $27,617$ Pion Tracks

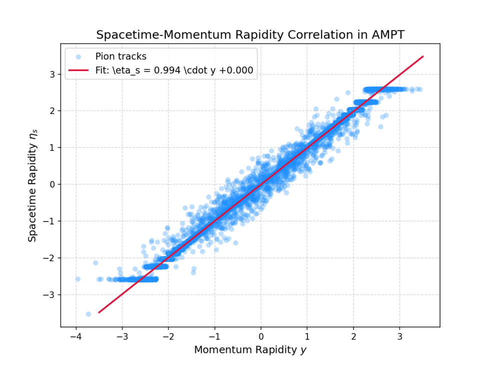
In Problem 5 of the laboratory, you map out the (y, η_s) scatter correlation for freezeout pions using real AMPT simulations.

- **Linear correlation r:** 0.9973
- **Fitted slope:** 0.9937

Physical Conclusion

This near-unity slope and high correlation coefficient represent a direct validation of the **Bjorken scaling model** of relativistic longitudinal expansion!

AMPT Spacetime-Momentum Correlation



3D Kinematics in Detector Planes

Transverse & Longitudinal Decomposition (Sahoo Fig. 5.8)

A particle's momentum $\vec{p}$ is decomposed relative to the beam axis (Z-axis) into: - **Transverse Momentum (p_T):** curvature in TPC: $p_T = \sqrt{p_x^2 + p_y^2}$ - **Transverse Mass (m_T):** $m_T = \sqrt{p_T^2 + m_0^2}$

Azimuthal Angle ϕ (Sahoo Eq. 5.50):

$$\phi = \tan^{-1}\left(\frac{p_y}{p_x}\right) \quad [-\pi, \pi]$$

Polar Angle θ (Sahoo Eq. 5.51):

$$\theta = \tan^{-1}\left(\frac{p_T}{p_z}\right) \quad [0, \pi]$$

Hyperbolic Mappings to Rapidity

Standard Rapidity y in term of m_T (Sahoo Eq. 5.54)

$$y = \frac{1}{2} \ln\left(\frac{E + p_z}{E - p_z}\right) = \ln\left(\frac{E + p_z}{m_T}\right)$$

Which highlights the hyperbolic boost rotation of coordinates:

$$E = m_T \cosh y, \quad p_z = m_T \sinh y$$

Pseudorapidity η (Sahoo Eq. 5.48)

For highly relativistic particles ($E \gg m_0 \Rightarrow m_T \approx p_T$), standard rapidity simplifies to a purely geometric quantity called **Pseudorapidity** (η), which depends only on polar angle θ:

$$\eta = \tanh^{-1}\left(\frac{p_z}{p}\right) = -\ln \tan\left(\frac{\theta}{2}\right)$$

Summary: Day 1 Hands-On Laboratory Exercises

Objective: Build custom Relativistic & Coordinate Calculators

All exercises are framed in your Jupyter Notebook at `exercises/day01_hands_on.ipynb`:

1. **Problem 1:** Relativistic velocity addition vs. rapidity additivity (Plot saturation).
2. **Problem 2:** 2-body CMS energy mappings for symmetric ($p + p$, $Au + Au$) and asymmetric ($p + Pb$) systems.
3. **Problem 3:** 3D Kinematics and coordinate decompositions (transverse variables and angles).
4. **Problem 4:** Light-cone momentum & rapidity on real AMPT data.
5. **Problem 5:** Spacetime rapidity mappings and correlation checks.